

Cartesian & Polar Coordinates Visualizer & Guide

Comprehensive Mathematical Guide with Visual Sketches & Solved Example 1

1. Fundamental Equations & Conversion Rules

Cartesian Coordinates: Represented by (x, y) coordinates on perpendicular axes.

Polar Coordinates: Represented by distance from origin (r) and directed angle (θ) as (r, θ).

Polar $\rightarrow$ Cartesian	Cartesian $\rightarrow$ Polar
$x = r \cdot \cos(\theta)$	$r = \sqrt{x^2 + y^2}$
$y = r \cdot \sin(\theta)$	$\theta = \arctan(y / x)$

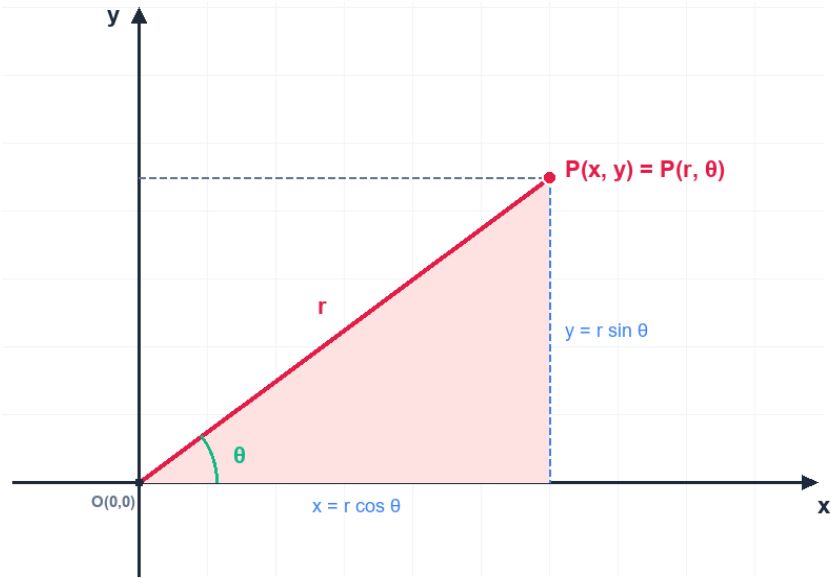


Figure 1: Geometric Relationship & Right-Triangle Projections for $P(x,y)$ and $P(r, \theta)$

2. Direction of Angles: Positive (+) vs Negative (-)

- **Positive Angle (+ θ):** Measured counter-clockwise from positive x-axis.
- **Negative Angle (- θ):** Measured clockwise from positive x-axis.

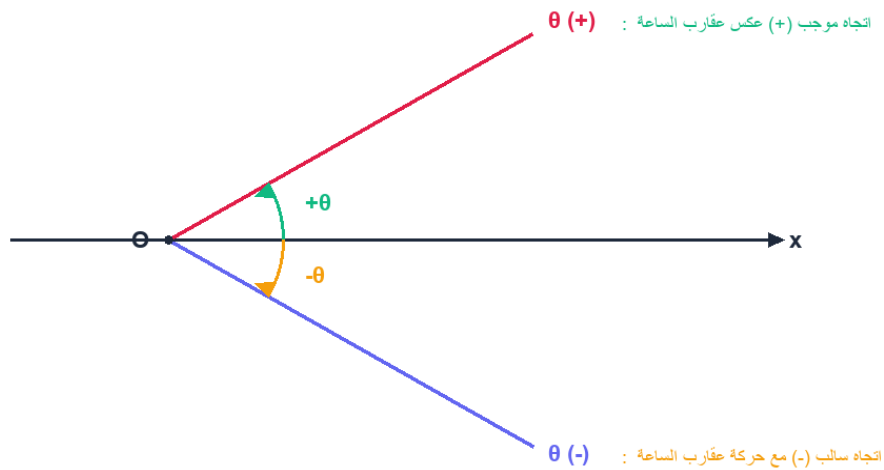


Figure 2: Positive angle direction (+) vs Negative angle direction (-)

3. Example 1: Sketching Points & Negative Radius Equivalence

Problem: Plot points $(1, 30^\circ)$, $(2, 30^\circ)$, $(-1, 30^\circ)$, $(1, 210^\circ)$, $(-1, -150^\circ)$.

Key Concept: When $r < 0$, travel in the opposite direction along the ray (add/subtract 180°).

Therefore: $(-1, 30^\circ) \equiv (1, 210^\circ) \equiv (-1, -150^\circ)$ — all land on the exact same location in Quadrant III.

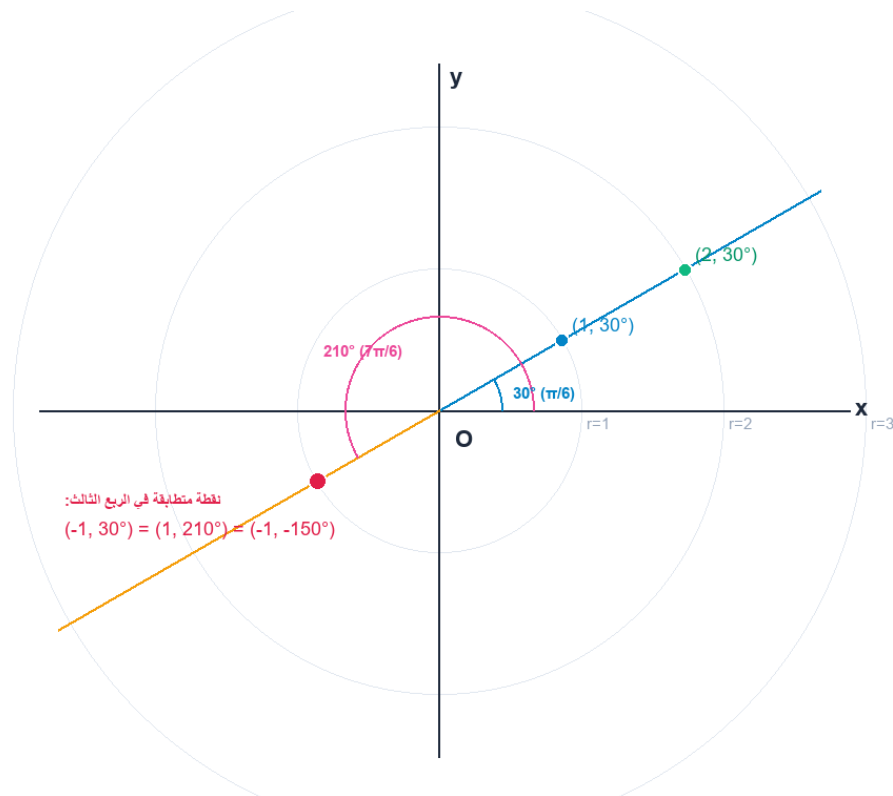


Figure 3: Graphical verification of Ex 1 points and equivalence in Quadrant III